

SAS: A Scalar Anonymous Communication System

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Abstract. Anonymity technologies have gained more and more attention for communication privacy. In general, users obtain anonymity at a certain cost in an anonymous communication system, which uses rerouting to increase the system's robustness. However, a long rerouting path incurs large overhead and decreases the quality of service (QoS). In this paper, we propose the Scalar Anonymity System (SAS) in order to provide a tradeoff between anonymity and cost for different users with different requirements. In SAS, by selecting the level of anonymity, a user obtains the corresponding anonymity and QoS and also sustains the corresponding load of traffic rerouting for other users. Our theoretical analysis and simulation experiments verify the effectiveness of SAS.

1 Introduction

In this paper, we propose a scalar anonymity system, which provides users a tradeoff between the degree of anonymity, and the cost.

Concerns about privacy and security have gained more and more attention in conjunction with the rapid growth and public acceptance of the Internet as a means of communication and information dissemination. Anonymity has become necessary and legitimate in many scenarios, including anonymous email, web browsing and e-Voting. In these scenarios, encryption alone cannot achieve the anonymity required by participants [1,2,3].

Since Chaum [7] proposed mix networks, researchers have developed various anonymity systems for different applications: message-based, high-latency systems for applications such as anonymous email [13,16] and flow-based, low-latency systems [4, 8, 9] for latency sensitive applications such as anonymous Web browsing or other TCP applications. In this paper, we focus on flow-based mix networks.

In a flow-based mix network, in addition to properly encrypting packets, we need to reroute them through a series of Mixes to achieve strong anonymity in the presence of compromised nodes and global timing attacks. While additional Mixes on the route of a path inherently increase anonymity, long rerouting paths have a negative effect on the QoS, such as end-to-end latency, or TCP throughput. Moreover, most anonymity systems utilize overlay protocols, further increasing the overhead since

two nodes may span a number of routers. Thus, the performance of TCP applications in anonymity systems is very sensitive to the length of the path in terms of anonymity system nodes.

In this paper we propose the Scalar Anonymity System (SAS) to provide a tradeoff between anonymity and QoS. Our contributions are summarized as follows:

1. SAS provides scalar anonymity for users. That is, based on the QoS requirements of their applications, users can tune to the required level of anonymity.
2. SAS provides scalar forwarding load for users. The forwarding load of a node is defined as this node's number of appearances on all rerouting paths [4, 5]. Users who need a higher level of anonymity experience higher forwarding load.
3. We compare the effectiveness of predecessor attacks [6] against SAS with those of comparable systems. Our simulations show that SAS is highly resilient to predecessor attacks.

The remainder of this paper is organized as follows: We introduce the related work in Section 2 and network model in Section 3. In Section 4, we propose SAS and discuss its performance. We evaluate SAS in Section 5 and conclude the paper in Section 6.

2 Related Work

For anonymous email applications, Chaum [7] proposed to use relay servers, i.e. Mixes, rerouting messages, which are encrypted by public keys of the Mixes. An encrypted message is analogous to an onion constructed by the sender, who sends the onion to the first Mix. By using its private key, the first Mix peels off the first layer, which is encrypted using the public key of the first Mix's public key. After receiving the second Mix's address, the first Mix sends the peeled onion. This process continues in this recursive way. The core part of the onion is the receiver's address and the real message, which is then sent to the receiver by the last Mix. Chaum also proposed the return address and digital pseudonyms for users to communicate with each other in an anonymous way.

Helsingius [12] implemented the first Internet anonymous remailer, which is a single application node that just replaces the original email's source address with the remailer's address. It has no reply function and is subject to all the attacks mentioned below. Eric Hughes and Hal Finney [23] built the cypherpunk remailer, a real distributed mix network with reply functions that uses PGP to encrypt and decrypt messages. The system is subjected to a global passive attack and replay attack to its reply mechanism. Gülcü and Tsudik [13] developed a relatively full-fledged anonymous email system, Babel. Their reply technique does not need the sender to remember the secret seed to decrypt the reply message, but it is subject to replay attack. They studied the threat from the trickle attack, a powerful active attack. Another defect of Babel is that a mix itself can differentiate the forwarding and reply messages. Cottrell [14] developed Mixmaster, which counters a global passive attack by using message padding and also counters trickle and flood attacks [13, 15] by using a pool batching strategy. Mixmaster does not have a reply function. Danezis, Dingleline and Mathewson [16] developed Mixminion. Although Mixminion still has

many problems, its design considers a relatively complete set of attacks that researchers have found [15, 17-21]. The authors suggest a list of research topics for future study.

Low-latency anonymous communication can be further divided into systems using core mix networks and peer-to-peer networks. In a system using a core mix network, users connect to a pool of Mixes, which provides anonymous communication, and users select a forwarding path through this core network to the receiver. Onion routing [8], Freedom [10], and Tor [9] belong to this category. In a system using a peer-to-peer network, every node in the network is a Mix, but can also be a sender and a receiver. Obviously, a peer-to-peer mix network can be very large and may provide better anonymity in the case when many participants use the anonymity service and enough traffic is generated around the network. Crowds [4], Tarzan [11] and P⁵ [22] belong to this category.

3 Network Model

In this section, we introduce the network model used in our study. Low-latency anonymous communication can use either core networks or peer-to-peer networks. In a system using a *core network*, users connect to a pool of special nodes providing anonymity service and select a forwarding path through this core network to the receiver. *Onion Routing* [8], *Tor* [9], and many others belong to this category. In a *peer-to-peer anonymity network*, every node contributes to the anonymity service, but it can also be a sender, a receiver, or both. Obviously, peer-to-peer mix networks scale well and provide better anonymity if the number of participants is large. *Crowds* [4], *Tarzan* [11], and many others belong to this category.

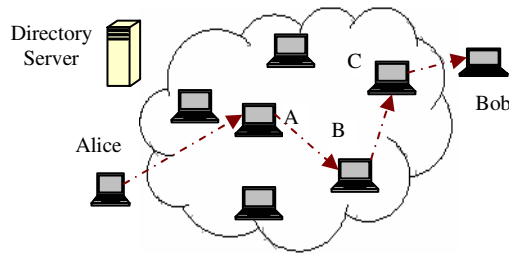


Fig. 1. Network Model

Figure 1 illustrates the network model. Here the Directory Server provides the information of nodes participating in the anonymity network¹. A sender Alice chooses nodes from the Directory Server and forms a path to the receiver Bob. For example, in Figure 1, Alice chooses relay nodes A, B and C and the path is Alice→A→B→C→Bob, where nodes A, B and C relay Alice's messages to Bob. Obviously, if the

¹ The directory server's function should be similar to the Directory Server used in Tor or Blender in Crowds.

Directory Server only stores a group of specified nodes, the whole anonymity system is core-network based and if the Directory Server stores every participating node, the whole anonymity system is peer-to-peer- network based.

The scalar anonymity system introduced in this paper is a general one and can use either core networks or peer-to-peer networks.

4 Scalar Anonymity System

In this section, we first introduce the framework of the Scalar Anonymity System (SAS), and then analyze its performance.

4.1 Framework of SAS

Rerouting-based anonymity systems typically rely on onion-like encryption against attacks by packet correlation or compromised mix nodes. In this paper, we will not provide elaboration. Instead, we will focus on the forwarding policy.

In this paper, we assume that the mix network adopts a forwarding policy as in Crowds [4], in which a probability guides if a packet is sent to the receiver or next hop. In order to provide a tradeoff between anonymity and QoS for different applications, each node selects a forwarding probability level. Assume that SAS provides m ($m \geq 1$) levels of anonymity corresponding to m different forwarding probabilities, denoted as the forwarding probability set $P_f = \{p_{f1}, \dots, p_{fk}, \dots, p_{fm}\}$. Without loss of generality we assume $p_{f1} < \dots < p_{fk} < \dots < p_{fm}$. Here we assume that a bigger forwarding probability provides stronger anonymity, and this assumption holds in our context although it may not hold in other cases [24].

In order to provide a tradeoff between anonymity and forwarding load, every node will be assigned a forwarding load level according to its anonymity level. Assume that SAS maintains m ($m \geq 1$) levels of forwarding load, denoted as the forwarding load level set $L = \{l_1, \dots, l_k, \dots, l_m\}$. Each node existing in the directory server will be assigned a weight equal to a forwarding load level from L . Assume that the numbers of nodes with forwarding load levels l_1 to l_m are $n_1, \dots, n_k, \dots, n_m$ and $n = n_1 + \dots + n_k + \dots + n_m$, where n is the total number of nodes in the system.

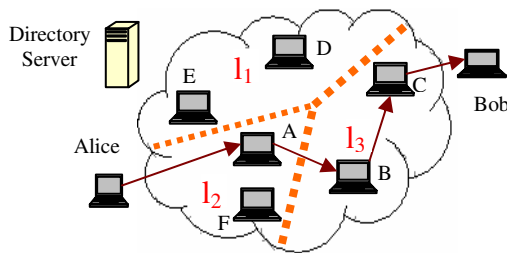


Fig. 2. Scalar Anonymity Systems

Figure 2 illustrates the weight assignment strategy. We have three forwarding load levels l_1, l_2, l_3 . Nodes E and D have forwarding load level l_1 , nodes A and F have forwarding load level l_2 , and nodes B and C have forwarding load level l_3 . Alice and Bob can also be assigned the forwarding load level if SAS is a peer-to-peer network based system.

The protocol works as follows.

1. Alice selects a forwarding probability p_{fk} , which is assigned as the forwarding probability for her packets. The directory server assigns to this node, a forwarding load level according to its forwarding probability level, which will be used as the weight of her node.
2. Alice constructs the path using a non-fair coin. When Alice flips the coin, the head appears with a probability p_{fk} . Whenever Alice has a packet to send, she performs the following steps:
 - a. She flips the coin. If it turns up with tail facing, she sends the packet to Bob, otherwise, she selects a node to which she forwards the packet. In SAS, a node with weight l_k is chosen with the probability of $l_k / \sum_{i=1}^m l_i \cdot n_i$. Thus, the bigger the weight of a node, the higher probability that the corresponding node is chosen as Alice's next forwarding node.
 - b. Say Node A in Figure 2 is chosen as Alice's next forwarding node. Node A flips the same coin again and decides if it should forward the packet. This process repeats until one node like Node C in Figure 2 decides to forward the packet to Bob.
3. Once the path is chosen, Alice encrypts the packet in an onion-like way as in Tor and sends the packet out to her first forwarding node.

We claim that the above protocol can achieve anonymity, QoS, and forwarding load in a scalar fashion. Users can choose different forwarding probabilities, and so select a desired level of anonymity. On the other hand, users control the latency (and therefore the QoS) of flows through the forwarding parameter. Furthermore, users can experience different forwarding load by selecting different forwarding probabilities. In the following paragraphs, we verify our claims.

4.2 Anonymity Analysis of SAS

We study SAS's anonymity in terms of the predecessor attack [4,6]. We will show that users with a higher forwarding probability tend to be more resilient to the predecessor attack.

Overview of the Predecessor Attack. This attack is based on the assumption that a sender might choose to remain in contact with a receiver for an extended period of time. In that case, the session between the sender and receiver is subject to a number of resets, (i.e., Web browsing). The interval between two subsequent resets is called a round. A sender will construct a rerouting path in every round. We assume that the attacker compromises some nodes to assist him collecting useful information. In every round, if compromised nodes appear on the rerouting path, the attacker logs the direct predecessor of the first compromised node. After a certain number of rounds,

the attacker analyzes what he has logged and determines the sender, which is selected based on the biggest number of appearances. Please refer to [4,6] for details.

SAS' Resistance to Predecessor Attacks. Recall in SAS, we assume that the numbers of nodes are $n_1, \dots, n_k, \dots, n_m$ corresponding to forwarding load level $l_1, \dots, l_k, \dots, l_m$, and the total number of nodes is n . Assume that there are c_k compromised nodes in forwarding load level l_k , this leads to the following theorem.

Theorem 1. In SAS, the upper bound of rounds, R_k , that an attacker needs to determine the sender of forwarding probability p_{fk} , is given in (1).

$$R_k = \frac{n'}{c'} \log(n') + \frac{n'}{c' - n' \sigma'} \log(n') \quad (1)$$

where

$$n' = \sum_{i=1}^m l_i n_i, \quad (2)$$

$$c' = \sum_{i=1}^m l_i c_i, \quad (3)$$

and σ' is given in (4),

$$\sigma' = \frac{p_{fk} c'}{n'^2 - n' p_{fk} (n' - c')}. \quad (4)$$

We have the following observations from Theorem 1: R_k is an increasing function of the forwarding probability p_{fk} . To verify this, we substitute (4) into (1), and get (5) which clearly shows the relationship between R_k and p_{fk} .

$$R_k = \frac{n'}{c'} \log(n') \left(1 + \frac{1}{1 - \frac{n'}{\frac{n'^2}{p_{fk}} - n'(n' - c')}} \right) \quad (5)$$

Thus, an anonymity system can achieve scalar anonymity by letting users choose the forwarding probability p_{fk} .

4.3 Quality of Service in SAS

In this subsection, we will discuss the properties of QoS provided by SAS. Without loss of generality, QoS is measured by the delay which is measured by the length of the rerouting path. We will show that users are able to effectively trade off anonymity against QoS in SAS.

Lemma 1. The expected path length, $E(PathLength_k)$, of a node with forwarding probability p_{fk} can be calculated in (6).

$$E(PathLength_k) = \frac{1}{1 - p_{fk}} \quad (6)$$

From Lemma 1, we can see that, the expected path length is an increasing function of p_{fk} . Thus, an anonymity system like SAS can achieve scalar QoS by letting users choose the forwarding probability p_{fk} , i.e. users can tune the level of anonymity based on the QoS requirements of their applications.

4.4 Forwarding Load in SAS

Recall in SAS, we assume that the numbers of nodes are $n_1, \dots, n_k, \dots, n_m$ corresponding to forwarding load levels $l_1, \dots, l_k, \dots, l_m$. Assume that there are $P_1, P_2, \dots, P_k, \dots, P_m$ rerouting paths corresponding to the forwarding probability levels $p_{f1}, p_{f2}, \dots, p_{fk}, \dots, p_{fm}$. We have the following theorem.

Theorem 2. In SAS, the expected load, $E(Load_k)$, of a node with forwarding load level l_k is given in (7).

$$E(Load_k) = l_k \frac{\sum_{k'=1}^m \frac{P_{k'}}{(1 - p_{fk'})}}{n'} \quad (7)$$

Where, n' is defined in (2), and $P_{k'}$ is the number of rerouting paths initiated by nodes with forwarding probability $p_{fk'}$.

From Theorem 2, we can see that the expected load of a node with forwarding load level l_k , is an increasing function of l_k . Thus, an anonymity system like SAS can achieve scalar forwarding load.

5 Evaluation

In this section, we will evaluate SAS from three aspects: anonymity, QoS, and forwarding load. In our simulation we choose forwarding load levels $l_1=1, l_2=2, l_3=3$, and forwarding probability levels $p_{f1}=0.5, p_{f2}=0.7, p_{f3}=0.9$. The numbers of nodes corresponding to three forwarding load levels are $n_1=300, n_2=300, n_3=300$ and the numbers of rerouting paths corresponding to three forwarding probability levels are $P_1=300, P_2=300, P_3=300$. We assume that 1% of the nodes are compromised.

5.1 Anonymity in SAS

Figure 3 shows the simulation results of predecessor attacks determining a sender with different forwarding probability levels in SAS. Each data point is based on the average of thirty runs. The figure shows that if an attacker hopes to find the sender with probability 0.5, they need to observe about 54 rounds in order to find a sender with forwarding probability 0.5, need to observe about 160 rounds in order to find a sender with forwarding probability 0.7 and need to observe 220 rounds in order to find a sender with forwarding probability 0.9. That is to say, the bigger the forwarding probability is, the more rounds for the attacker need to observe.

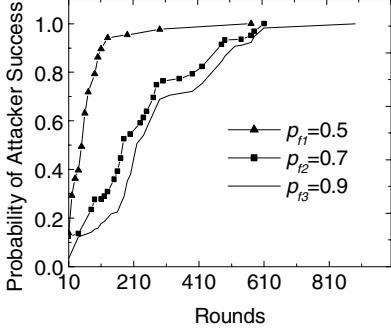


Fig. 3. Simulations of the predece-ssor attack against nodes with different forwarding probability levels in SAS

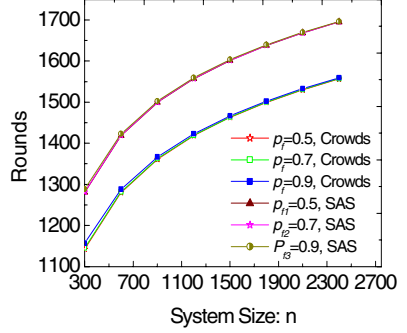


Fig. 4. Comparisons of rounds for predece-ssor attacks in SAS and in Crowds

Figure 4 shows the comparison results of the round upper bound for an attacker to determine a sender in SAS and in Crowds with high probability. The figure shows that an attacker requires many more rounds to find the sender with high probability in SAS than in Crowds when the compromised nodes are distributed uniformly. For example, in a Crowds system with 600 nodes and a forwarding probability of 0.9, an attacker only needs to observe 1288.312 rounds to determine the sender with high probability when the rate of compromised nodes is 1%. However, In a SAS with the same network, an attacker needs to observe 1422.928 rounds to determine the sender, whose forwarding probability is 0.9 when the rate of compromised nodes is 1%.

5.2 QoS and Forwarding Load in SAS

In SAS, we achieve the tradeoff between QoS and anonymity by controlling the path length and the tradeoff between forwarding load and anonymity by using the forwarding load level.

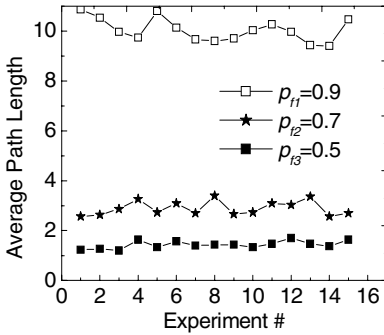


Fig. 5. Simulation of path length for nodes with different forwarding probability levels in SAS

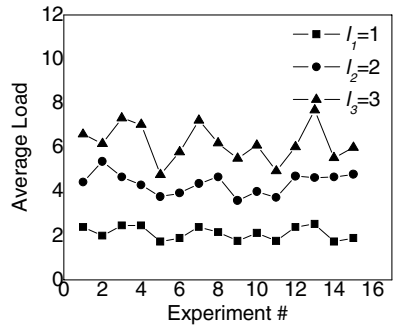


Fig. 6. Simulations of load for nodes with different forwarding load levels in SAS

Figure 5 shows the average path length under different forwarding probabilities. The horizontal axis is the index of our experiments. We carried out 15 groups of simulation. The figure shows that, the bigger the forwarding probability, the longer the rerouting path, hence the longer the delay. Figure 6 shows the simulation results of forwarding load for nodes with different forwarding load levels. Each data point is based on the average forwarding load of all nodes with the same forwarding load levels. The figure shows that, the bigger the forwarding load level, the heavier the forwarding load.

6 Conclusion

In this paper, we propose a scalar anonymous communication system. The scalar anonymity is achieved with layered encryption, and multi-hop routing. In SAS, we achieve a tradeoff among QoS, forwarding load, and anonymity by providing different forwarding probability levels for different nodes, and by assigning a forwarding load level for every node. We demonstrated the effectiveness of SAS by theoretical analyses and simulations. Moreover, we calculated the rounds that an attacker deploying predecessor attack needs in order to determine a sender with high probability in SAS, and compared SAS with Crowds under the same attack. We find that SAS is more resilient to this attack than Crowds when the compromised nodes are uniformly distributed.

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